Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

typearea 15 algorithm2e natbib latexsym,graphicx amsmath,amssymb,enumerate amsthm graphicx float xspace bm pifontifpdf shadow,shadethm,color [compact]titlesec listings color

lipsum xargs [pdftex,dvipsnames]xcolor colortbl carnationpinkrgb1.0, 0.65, 0.79 Graygray0.9 [first=0,last=9]lcg

[colorinlistoftodos,prependcaption,textsize=tiny]todonotes [2][1=][linecolor=red,backgroundcolor=red!25,bordercolor
[2][1=][linecolor=blue,backgroundcolor=blue!25,bordercolor=blue,1]2 [2][1=][linecolor=OliveGreen,backgroundcolor=Ol

[2][1=][linecolor=Plum,backgroundcolor=Plum!25,bordercolor=Plum,1]2 [2][1=][disable,1]2 *arg max *arg min

codegreenrgb0,0.6,0 codegrayrgb0.35,0.35,0.35 codepurplergb0.58,0,0.82 backcolourrgb0.95,0.95,0.92

mystyle backgroundcolor=backcolour, commentstyle=codegreen, keywordstyle=magenta, numberstyle=codegray, stringstyle=codepurple, basicstyle=, breakatwhitespace=false, breaklines=true, captionpos=b, keepspaces=true, numbers=left, numbersep=5pt, showspaces=false, showstringspaces=false, showtabs=false, tabsize=2

style=mystyle

subfigure verbatim paralist fancyvrb

enumerate enumitemchangepage

Darkbluergb0,0,0.4 Browncmyk0,0.81,1.,0.60 Purplecmyk0.45,0.86,0,0 [breaklinks,pdftex]hyperref

*argmin

19 Pebruari 2025

PROSPERO Application Form

Asterisk $*$ indicates mandatory fields.

1 *Review title

codegrayGive the title of the review in English

2 Original language title.

codegrayFor reviews in languages other than English, give the title in the original language. This will be displayed with the English language title.

3 *Anticipated or actual start date.

codegrayGive the date the systematic review started or is expected to start.

4 *Anticipated completion date.

codegrayGive the date by which the review is expected to be completed.

5 * Stage of review at time of this submission.

codegray Tick the boxes to show which review tasks have been started and which have been completed. Update this field each time any amendments are made to a published record. Reviews that have started data extraction (at the time of initial submission) are not eligible for inclusion in PROSPERO. If there is later evidence that incorrect status and/or completion date has been supplied, the published PROSPERO record will be marked as retracted. This field uses answers to initial screening questions. It cannot be edited until after registration.

Stage	Started	Completed
Preliminary searches:		
Piloting of the study selection process:		
Formal screening of search results against eligibility criteria:		
Data extraction:		
Risk of bias (quality) assessment:		
Data analysis:		

6 *Named contact.

codegrayThe named contact is the guarantor for the accuracy of the information in the register record. This may be any member of the review team.

Email salutation (e.g. "Dr Smith" or "Joanne") for correspondence:

7 *Named contact email.

codegrayGive the electronic email address of the named contact.

8 Named contact address.

codegrayGive the full institutional/organisational postal address for the named contact.

9 Named contact phone number.

codegrayGive the telephone number for the named contact, including international dialling code.

10 *Organisational affiliation of the review.

codegrayFull title of the organisational affiliations for this review and website address if available. This field may be completed as 'None' if the review is not affiliated to any organisation.

Organisation web address:

11 *Review team members and their organisational affiliations.

codegrayGive the personal details and the organisational affiliations of each member of the review team. Affiliation refers to groups or organisations to which review team members belong. NOTE: email and country now MUST be entered for each person, unless you are amending a published record.

12 *Funding sources/sponsors.

codegrayDetails of the individuals, organizations, groups, companies or other legal entities who have funded or sponsored the review.

Grant number(s)

codegrayState the funder, grant or award number and the date of award

13 *Conflicts of interest.

codegrayList actual or perceived conflicts of interest (financial or academic).

14 Collaborators

codegrayGive the name and affiliation of any individuals or organisations who are working on the review but who are not listed as review team members. NOTE: email and country must be completed for each person, unless you are amending a published record.

15 *Review question.

codegrayState the review question(s) clearly and precisely. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS or similar where relevant.

16 *Searches.

codegrayState the sources that will be searched (e.g. Medline). Give the search dates, and any restrictions (e.g. language or publication date). Do NOT enter the full search strategy (it may be provided as a link or attachment below.)

17 URL to search strategy.

codegrayUpload a file with your search strategy, or an example of a search strategy for a specific database, (including the keywords) in pdf or word format. In doing so you are consenting to the file being made publicly accessible. Or provide a URL or link to the strategy. Do NOT provide links to your search results. Alternatively, upload your search strategy to CRD in pdf format. Please note that by doing so you are consenting to the file being made publicly accessibl

18 *Condition or domain being studied.

codegrayGive a short description of the disease, condition or healthcare domain being studied in your systematic review.

19 * Participants/population.

codegraySpecify the participants or populations being studied in the review. The preferred format includes details of both inclusion and exclusion criteria.

20 *Intervention(s), exposure(s).

codegrayGive full and clear descriptions or definitions of the interventions or the exposures to be reviewed. The preferred format includes details of both inclusion and exclusion criteria.

21 *Comparator(s)/control.

codegrayWhere relevant, give details of the alternatives against which the intervention/exposure will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

22 *Types of study to be included.

codegrayGive details of the study designs (e.g. RCT) that are eligible for inclusion in the review. The preferred format includes both inclusion and exclusion criteria. If there are no restrictions on the types of study, this should be stated.

23 Context.

codegrayGive summary details of the setting or other relevant characteristics, which help define the inclusion or exclusion criteria.

24 Main outcome(s).

codegrayGive the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurement are made, if these are part of the review inclusion criteria.

*Measures of effect

codegrayPlease specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

25 *Additional outcome(s).

codegrayList the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate to the review

* Measures of effect

codegrayPlease specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

26 *Data extraction (selection and coding).

codegrayDescribe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

27 *Risk of bias (quality) assessment.

codegrayState which characteristics of the studies will be assessed and/or any formal risk of bias/quality assessment tools that will be used.

28 Strategy for data synthesis.

codegrayDescribe the methods you plan to use to synthesise data. This must not be generic text but should be specific to your review and describe how the proposed approach will be applied to your data. If metaanalysis is planned, describe the models to be used, methods to explore statistical heterogeneity, and software package to be used.

29 *Analysis of subgroups or subsets.

codegrayState any planned investigation of 'subgroups'. Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.

30 *Type and method of review.

codegraySelect the type of review, review method and health area from the lists below.

Type of review

Cost effectiveness	Diagnostic	Epidemiologic
Individual patient data (IPD) meta-analysis	Intervention	Meta-analysis
Methodology	Narrative synthesis	Network meta-analysis
Pre-clinical	Prevention	Prognostic
Prospective meta-analysis (PMA)	Review of reviews	Service delivery
Synthesis of qualitative studies	Systematic review	Other

Health area of the review

Alcohol/substance misuse/abuse	Blood and immune system	Cancer
Cardiovascular	Care of the elderly	Child health
Complementary therapies	COVID-19	Crime and justice
Dental	Digestive system	Ear, nose and throat
Education	Endocrine and metabolic disorders	Eye disorders
General interest	Genetics	Health inequalities/health equity
Infections and infestations	International development	Mental health and behavioural conditions
Musculoskeletal	Neurological	Nursing
Obstetrics and gynaecology	Oral health	Palliative care
Perioperative care	Physiotherapy	Pregnancy and childbirth
Public health (including social determinants of health)	Rehabilitation	Respiratory disorders
Service delivery	Skin disorders	Social care
Surgery	Tropical Medicine	Urological
Wounds, injuries and accidents	Violence and abuse	

31 Language

codegraySelect each language individually to add it to the list below, use the bin icon to remove any added in error.

32 *Country.

codegraySelect the country in which the review is being carried out. For multi-national collaborations select all the countries involved.

33 Other registration details.

codegrayName any other organisation where the systematic review title or protocol is registered (e.g. Campbell, or The Joanna Briggs Institute) together with any unique identification number assigned by them. If extracted data will be stored and made available through a repository such as the Systematic Review Data Repository (SRDR), details and a link should be included here. If none, leave blank.

34 Reference and/or URL for published protocol.

codegrayIf the protocol for this review is published provide details (authors, title and journal details, preferably in Vancouver format) Add web link to the published protocol. Or, upload your published protocol here in pdf format. Note that the upload will be publicly accessible. Please note that the information required in the PROSPERO registration form must be completed in full even if access to a protocol is given.

35 Dissemination plans.

codegrayDo you intend to publish the review on completion?
codegrayGive brief details of plans for communicating review findings?

36 Keywords

codegrayGive words or phrases that best describe the review. Separate keywords with a semicolon or new line. Keywords help PROSPERO users find your review (keywords do not appear in the public record but are included in searches). Be as specific and precise as possible. Avoid acronyms and abbreviations unless these are in wide use.

37 Details of any existing review of the same topic by the same authors.

codegrayIf you are registering an update of an existing review give details of the earlier versions and include a full bibliographic reference, if available.

38 *Current review status.

codegrayUpdate review status when the review is completed and when it is published.New registrations must be ongoing so this field is not editable for initial submission. Please provide anticipated publication date

39 Any additional information.

codegrayProvide any other information relevant to the registration of this review.

40 Details of final report/publication(s) or preprints if available.

codegrayLeave empty until publication details are available OR you have a link to a preprint (NOTE: this field is not editable for initial submission). List authors, title and journal details preferably in Vancouver format. Give the link to the published review or preprint.